

Math 242, S16
Quiz 12

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Use Laplace transform to solve the initial value problem

$$x'' + 9x = 4t; \quad x(0) = x'(0) = 0.$$

Solution: Let $\mathcal{L}\{x(t)\} = X(s)$, then

$$\mathcal{L}\{x'(t)\} = sX(s), \quad \mathcal{L}\{x''(t)\} = s^2X(s).$$

Thus

$$\mathcal{L}\{x'' + 9x\} = \mathcal{L}\{4t\}$$

$$\mathcal{L}\{x''\} + 9\mathcal{L}\{x\} = 4 \frac{1}{s^2}$$

$$+ 9X(s) = \frac{4}{s^2}$$

$$(s^2 + 9)X(s) = \frac{4}{s^2}$$

$$X(s) = \frac{4}{s^2(s^2 + 9)}$$

Check

$$\frac{4}{s^2(s^2 + 9)} = \frac{A}{s^2} + \frac{B}{s^2 + 9} = \frac{A(s^2 + 9) + Bs^2}{s^2(s^2 + 9)}$$

$$= \frac{(A+B)s^2 + 9A}{s^2(s^2 + 9)}$$

$$\Rightarrow \begin{cases} A+B=0 \\ 9A=4 \end{cases} \Rightarrow \begin{matrix} A = \frac{4}{9} \\ B = -\frac{4}{9} \end{matrix}$$

Thus

$$\begin{aligned} x(t) &= \mathcal{L}^{-1}\left\{\frac{4}{s^2(s^2 + 9)}\right\} \\ &= \mathcal{L}^{-1}\left\{\frac{4}{9} \frac{1}{s^2} - \frac{4}{9} \frac{1}{s^2 + 9}\right\} \\ &= \frac{4}{9}t - \frac{4}{27}\sin 3t. \end{aligned}$$

Problem 2. (4 points) Use Theorem of Differential Transforms (i.e., $\mathcal{L}\{tf(t)\} = -F'(s)$) where $F(s) = \mathcal{L}\{f(t)\}$ to compute $\mathcal{L}\{t \sin 4t\}$.

Solution: Let $f(t) = \sin 4t \Rightarrow F(s) = \mathcal{L}\{\sin 4t\} = \frac{4}{s^2 + 16}$

Thus

$$\mathcal{L}\{t \sin 4t\} = -\frac{d}{ds} \left[\frac{4}{s^2 + 16} \right]$$

$$= -4(-1)(s^2 + 16)^{-2} \cdot (2s)$$

$$= \frac{8s}{(s^2 + 16)^2}$$